

ActInf MorphStream 003.1 ~ Anand Subramoney "Principles of scalability a

MorphStream | Anand Subramoney | 1:01:30

[Laughter]

so hello everybody oh can you just go on
mute if you're not speaking sorry I'm
just going to do a bit of an
introduction um so welcome to the third
um morph stream series the others are on
YouTube if you've not had a chance to
catch those so just a reminder of the
purpose of these meetings are to educate
and Inspire at the intersection of
active inference and neuromorphic
Computing um if you've not met me before
I'm Sarah Hamburg so I'm predominantly a
neuroscientist I did my PhD neuroscience
at UCR which is where I first came
across active inference um and I've just
finished a postto in neuromoritic
Computing and I'm part of the Sab for
the active inference Institute and I've
think I've been involved with them since
about 2020 when I was trying to work out
earlier but I'm really pleased today
that we're joined by Professor Annan sub
money so he's an assistant professor in
the Department of computer science at
Royal Holloway London he's broadly
interested in learning and intelligence
so that's both algorithmic and
biological his current research is
focusing on understanding the principles
of scalability in deep learning which is
what he'll be talking about today sorry
I'm just admitting some people um so he
aims to use principles of scalability to
build models that are more efficient and

can scale up seamlessly his research draws inspiration from neuroscience and biology in his Quest To Build a Better and more General artificial intelligence and you can visit his website on and subber money for more information I'll pop that in the chat as well so his talk today is on principles of scalability and biological Inspirations he's going to talk about how current models scale and what we can learn from the efficiency of biological brains one of the central themes will be sparsity its significant role in scalable systems and its synergies with neuromorphic Hardware he'll present existing ideas based on spiking networks and recent work from his group which is focused on using various forms of sparsity and distributed learning to improve the scalability and efficiency of our learning models so the format will be that the first part um arnand will speak and then we'll sort of have a more General open conversation between all of us um arand I don't know if you want people to in interrupt you with questions or not if you just want to take them at the end that's entirely up to you yeah I'm actually completely fine with taking questions during the talk cool okay right well without further Ado uh yeah we'd love to hear your presentation yeah thank you Sarah and thank you very much for the invitation uh so yeah like Sarah mentioned I'm going to be talking about principles of scalability and biological

Inspirations uh so I've been working with computation neuroscience and machine learning for quite a long time and have been thinking about uh well I mean how intelligence Works to some extent but also really uh a big part of it is about understanding how various how biological models scale and by extension how artificial intelligence model scale and that's what I'm going to be talking about today so just to start off with uh things that we already know I mean so AI is really starting to become better than humans at lots of things right so this is kind of like a a plot of performance of various AI models in relation to humans and you see that handwriting recognition and speech recognition image recognition so this all of these have been like improving very very fast uh but recently things like language understanding and reading comparison have also in imp improved and become better than humans so there is really some kind of like a major movement going on in AI where things are really becoming uh better very fast and so the question is what is happening or why does it so work so well so there have been a lot of uh advances in the past do decades uh because we've started using deep neural networks with lots of layers and lots of parameters and we have uh as as a the machine Learning Community has figured out how to train these models well and get them to work well so there has been

algorithmic advances in stochastic
gradient descent using Rao layer skip
connections normalization layers and all
of these have been like very important
in
giving us the ability to train these
very deep
networks the other aspect uh that's
things the other thing that has changed
in the past two decades is that we have
lots more data right so we have access
to a lot more data so uh which helps a
lot with supervised learning but then
we've also uh as a community figured out
how self-supervised learning can be used
to train these models and get them to do
interesting
things and the third thing is we have a
lot more compute and I will talk about
this so this is uh actually going to be
one of the focus of my
talk uh and we also have scalable
architectures right so even if you have
lots of compute it doesn't really help
if uh the machine learning architectures
that we have don't scale and make use of
the compute that we have so that's
obviously like a very closely related
and important part so just to look at
compute itself so this is kind of like a
plot of various models all the way from
1985 to modern day and so what's
interesting here is I mean first of all
uh before GPU Computing which is this
left side of the plot over
here uh what you see is that um the
computer has been doubling every 24
months so this has been just standard M

law we' have been getting faster and faster processes but there is this sort of discontinuity where people figured out that they could use gpus that were originally designed for for graphics uh they could use that to do machine learning and so what you see here is that the slope of the uh increase in per in amount of compute available changes quite significantly it becomes a lot more

Steep and so here the compute starts doubling every 3.4 months starting about 2012 or

so and then the third major aspect is these set of models uh that we're familiar with now the Transformer models and you see that uh the comput has been doubling every two months uh with with these models so uh so it looks like compute obviously plays a very very major role in the AI capabilities that we have but also the models that are able to take advantage of this computer so what happened since 2018 which is what uh uh which is when uh this comput started doubling every two months so one of the major advances in 2018 was uh a new scalable deep learning architectures that were published uh that actually was able to take use of this or make use of the compute very well and then a lot of engineering effort and um money spent uh actually getting these models uh scaling up these models built on these architectures right actually making use of the computer and this is

kind of uh always a separate concern because even if you have a great model that scales really well there needs to be like a lot of engineering done to actually scale it

up and the scalable model that was published in 2018 was the Transformers so most of you might have seen this

already so this is the paper attention is all you need which was published in two first published in

2017 and has already gotten tens of thousands of citations if not more close to 100,000 citations and uh this is an architecture that replaced lstms which were the dominant recurrent Network architecture for sequence processing at the

time so

um so there are various reasons why Transformers are scalable uh and what was kind of interesting is

that so they're scalable not just in the ability to be able to run them across a lot of gpus or to use up a lot of compute but uh the important aspect of scaling up over here is that as you spent more compute uh the model performance

improved right and that's that's a really important aspect and this is kind of the scaling plot from the now famous uh gpt3 paper and what you see is uh on the x-axis is the amount of compute that was used to train the model so this kind of includes a combination of the model size and the uh length of the training

which is also like very closely related to the amount of data that was available and uh the y axis here is the validation error and and what you see here is that as you spend more and more compute as you have larger models and as you train them for longer uh with with certain amounts of data the validation error goes down and uh so uh you can you can really like invest comp and get returns out of using using a poet comput and here there's like a power law relationship between these two and depending upon the model more recently people have observed there's like a much more complicated relationship but these models do tend to improve in performance as you invest more compute in them uh up till up to a certain limit obviously okay so why do these uh architecture scale well is I mean obviously one of the main questions and then by extension can we scale these models even better and how do we do this so to answer the question of why do Transformers scale so well I mean there are lots of reasons I'm just like focusing on a few of them but uh like I said Transformers were uh proposed as a replacement to recurrent networks in the form of lstms and they use something called self attention to process a quences so U what they did was they replaced recurrent networks which which are known to be hard to train with feed forward networks that are easier to train and uh the

other kind of very important U advantage of Transformers was that they allow parallelization across the length of the sequence right and so when you have like really long sequences or I mean any number of sequences in any length of sequence the recur Network you always have to uh process it one SE sequence element at a time and with Transformers you could process the entire length of a sequence in one go because of the particular way the architecture is designed and that's a pretty huge Advantage because now you have a whole new dimension of parallelization and um so the second reason why Transformers scale so well is because uh we already figured out that gpus work well for machine learning and Transformers were really really uh great fit for gpus right so you could a Transformer and a GPU and really make full use of the compute provided by the GPU I mean we don't know if uh Transformers are intrinsically better as a model or if it is just because of the good fit between Transformers and the available compute U the available types of compute that we have today so gpus primarily so uh which is which is kind of this uh question of was it a hardware Lottery right um so a hardware Lottery is used to describe when a research idea wins because it is suited to the available software and Hardware better and not because the idea is superior to the alternative research directions and so for Transformers it's not fully clear

if it was just a hardware lottery or if there's like something uh very intrinsically advantageous about it okay

so so so the scaling of Transformers I mean I'm not going to go into any further Det because there's a lot of uh literature published on this but now the question is can we scale even better than Transformers and this is kind of a plot U uh taken from this paper by uh boan who's a professor in neuromorphic professor in trans

Stanford and uh so this is a plot of the uh on the x-axis you see the number of neurons and y axis the energy consumption right and uh so this Gray Line over here is where we are at currently so that's kind of the standard silicon chips uh in

2D and U the biological uh systems that we know of so biological brains is this darker line over here which kind of uh you see has a much steeper slope of scaling compared to the Silicon chips that we have so I'll talk about what this line in

between is in a moment but uh because of just like very practical reasons just physical constraints and the sizes of things and the amount of energy that it can consume and just the proportion of things like proportion of wiring uh versus the amount of compute that you have available so for various different reasons you really want the steeper slope uh in both the algorithms and the hardware that you have because

otherwise it makes it like close to impossible to achieve the same level of scale so over here biological brains have reached like 86 billion neurons and uh well I mean this is a little bit uh older but this because of the steep slope if you wanted to reach the same number of units you're going to consume quite a bit more energy right so Transformers are probably like somewhere up there and biological brains are over here and at some point this becomes a deal breaker so you can this this starts like uh providing like an uh upper limit on how big you can scale the models so Energy Efficiency in that sense is like uh like a very fundamentally important uh aspect u i mean assuming that scaling models always gives you this U dividend of better performance then uh it it at some point without efficient models it becomes impossible to reach sizes that can do interesting things

okay so uh what does biology do how does biology scale because we see that it has this steep slope of scaling and one of the things that it does is um it uses a lot of sparsity I mean a lot of it is also again like just because of physical limits and the amount of uh cells the density of the cells that you can have the amount of energy available to the brain and so on but mean since this has evolved under these constraints what we have ended up with are uh is a substrate for computation that

is extremely sparse both in time and space so uh this is kind of like a plot of What U the neuron activity looks like uh for for like one of these typical neurons and what you usually see is that there are these kind of spikes so there are times when the neuron decides to communicate with the rest of the neurons and then most of the rest of the time it's really not doing much and then so you have like these spikes of outputs and this is sparsity in time so um in terms of activity and then you also have sparsity in terms of space which is just the connectivity of the neuron is very very sparse so not all neurons in the brain are connected to all the other neurons so second thing is uh neurons uh use event based communication uh which means that communication really happens only through discrete events that send messages between units and so it's not the case that each neuron is talking with the next neuron all the time right which is uh on the other hand most of our artificial neural networks really do work like that so there's like an output that goes to the next unit or the connected units pretty much all the time uh the third thing and again this is because of just uh very practical constraints in biology is that learning is local in time and space so you cannot uh have uh altw communication between like neurons in the brain to do learning because it's just impractical and uh so what this means is

that each synapse has to use information

that's

available mostly locally to the synapse

and then with like very very low

dimensional Global signals incorporated

into this so uh and and and so this this

is par in uh is it's local in time as

well and the fourth aspect is uh temp

oral coding so because time is available

in some sense for free quote

unquote uh neurons can use time to

compute they can use time to actually

send

messages uh and it can be used in

various different ways uh so you can

neurons are able to trade off

computation for time in that sense there

are lots of other interesting properties

asynchrony for example uh and then also

some sort of like distributed learning

that happens in various parts of the

brain so I won't be talk going into

detail into any of those at the

moment okay so let's come back to this

plot which showed the scaling of various

things and so now I've added back the

labels from the original plot and so uh

this Gray Line over here is the dense 2D

silicon scaling that we have and so this

middle line over here is what would

happen if you actually took the silicon

and put it in three dimensions but

without sparsity and so you you obviously

do get like better scaling but there are

some practical challenges that need to

be overcome to actually have

three-dimensional silicon uh mostly

related to heating

issues and uh the biological brains are all in this regime of sparse three-dimensional um substrates right so this sparsity is really what makes the difference uh I mean given you have a three dimensional substrate sparsity is really what makes a difference in uh allowing this much better scaling property of these U computational devices as it were so this really means that we need uh apart from uh the right algorithms we also need the right hardware and uh there has been quite a lot of work on biologically inspired Hardware that can make use of the SP asynchronous communicating neural network units uh so uh and this is kind of like this illustration uh that Intel has so they they've been like also working on one of the uh neuromorphic Pro processors called lli and uh this is like a really great illustration of like the differences between conventional Computing and neuromorphic Computing but essentially in conventional Computing you have synchronous clocking you have sequential threads of control whereas in neuromorphic Computing you have completely asynchronous even based spikes and parallel and sparse compute and uh it's also I mean so gpus are kind of in between so which allows parallel Computing but it's still synchronous clocking and it's kind of kind of more optimized for dense computation and uh so these neuromorphic chips are essentially chips that use

these biological um ideas and are able to work with sparsity in both time and space and also able to work with asynchronous computation and so the state of the neuromorphic chips right now just to kind of uh summarize uh what's what's been happening so I mean I I personally don't work on the hardware aspect of it but mostly in uh the algorithms but there has been recable amount of progress with neuromorphic chips so for example um in 2021 at least uh so this is like a plot which shows U the uh relative performance of various uh devices in comparison with this Intel Loi chip which is a neuromorphic device and uh so if a device is if a point is above this dotted line it means that um Loi the neuromorphic device does better and if it's below it means the conventional device does better and all of the points here are conventional devices so you have a CP you have CPUs gpus um some kind of like accelerator deep learning accelerator and IBM True North which is also a neuromorphic device but a little bit older and uh the bat size also makes a huge difference if you have bat size of one the neuromorphic devices tend to do better in general and larger bat sizes uh gpus T to do better and so uh so so what you see over here is uh well first of all uh there are certain like biologically inspired models which are these blue ones over here that do much

better on a neuromorphic device compared to uh conventional

Hardware on the other hand uh what you see is that even for some of these biologically

inspired uh models if you start using large patch sizes so if you look at this purple Square over here uh it uses something called an LSN which was one of the uh papers that I was involved in uh a few years ago and if you combine that combine that with deep neural networks for a bat size of one uh the neuromorphic device does better but then if you improve increase the batch sizes then uh gpus suddenly start doing better so uh so neuromorphic devices are still not there yet it's still work in progress and uh Loi for example is a digital neuromorphic device and there's lot of uh variety of possible neuromorphic devices out there a lot of people working on very cool interesting ideas uh but there is this promise of uh uh the neuromorphic devices having really intrinsically better scaling curves if the engineering is done right okay so

now uh I'm going to talk about algorithms a little bit so let's just go back to this slide uh where we looked at what biology does uh that uh I think is important for scaling up these models and uh one of the initial kind of attempts at using some of these principles was what are called spiking neural networks and uh spiking neural networks

are essentially a direct biological
uh I mean it's it's a model of a neuron
that is really directly uh derived from
biology so as I said in biology you have
neurons and then if you kind of stick
electrone in the neuron and measure
um uh measure what it's doing uh what
you see is that it has this action
potential right so it's suddenly there
is this surge uh this the sudden change
in like the potential and I'll tell you
what this potential is in a minute and
uh but this uh action potential as it's
called is a spike and uh that's what is
used for communicating and most of the
rest of the time the neuron doesn't do
much and so this action potential is
essentially uh because the neuron is
constructed in such a way that there is
a voltage potential difference between
the inside and outside of the neuron and
most of the U and this potential is kind
of uh constructed through uh the
movement of ions the neurons control the
movement of ions but across the membrane
across the cell membrane and which kind
of creates a membrane potential and this
is used to kind of trigger the action
potential of a neuron which leads to
ionic currents inside the neuron which
teaches the synapsis and then there's a
chemical messaging across the synapse
which goes to the next neuron and so on
so that's kind of like the biological
process and this can be simplified into
like like a circuit uh if you will an RC
circuit and then it can further be
simplified into a computational model

where uh there are in inputs coming in as spikes which are just zero or one values and the neuron does a weighted sum of all the inputs coming in so this is kind of very similar to what artificial neural networks do except that in the case of spiking neural networks the inputs are binary and very very sparse in and uh so the neuron kind of integrates this and then if the internal state of the neuron goes above certain threshold then it also emits a sparse discrete Spike and uh so if you uh kind of uh look at the activity of the neuron this is what it looks like so the membrane potential if if there is a input that looks like this so some sort of uh constant input then the membrane potential goes up every time a SP Pike arrives at the cell body so in this case just the neuron the computational neuron and whenever this membrane potential goes above a certain threshold then the neuron emits a spike right okay so this is a very very simplified model and this model is called The Leaky integrated and fire model U so you can kind of write this down in terms of equations and essentially uh uh okay I mean so I I don't think I will go through this but um maybe the important thing to kind of note is that uh there is some sort of uh the output of the neuron which is Z over here is one if the me internal membrane potential is above thresold and zero if it is not and this Zed is the only thing

that other neurons see so the membrane potential is completely internal to this neuron and no other part of the model has access to it uh but this one zero value this binary value the spike that's emitted is what uh the other neurons see and this this kind of depends on the threshold okay

so now uh spiking neurons are great because you have this temporal sparsity now right so you're not like sending messages all the time but you're only sending messages every now and then but uh you need to train these models and in a standard RNN you can just use back propagation Through Time by defining a loss function and then calculating the gradients of the loss function back propagate It Through Time into the model and then update the the weights based on this gradient but the problem with spiking neural networks is that the spike emission function is non-differentiable which means that it's not so straightforward to just apply back propagation Through Time directly because that depends on all the parts of the model being differentiable but there is kind of like this uh very crude trick as it were that works surprisingly well uh that we discovered uh back back in 2018 along with colleagues which is that you can I mean so this uh blue curve over here is the spike emission function which is this kind of thres uh step function which where the derivative is not defined at the

threshold and uh we kind of realized and
I mean a lot of people realized at
around the same time that you could just
Define a pseudo derivative so you could
Define a derivative at that point quite
AR

Lily and uh within within some
constraints if you uh so for example we
uh defined it in this uh P linear
function and uh it tends to work really
really well so you just replace the
derivatives of this uh non
differentiable function at that
particular point with the value that's
derived from the pseudo derivative and
then you can just train all these models
with back propagation through time and
they work work really great but turns
out that that's not enough I mean so if
you have spiking neuron and use back
propagation through time uh that's
really not enough to get the same level
of performance as artificial neur
networks because the second aspect of
recurrent networks that we needed was
long short-term memory and recurrent
networks uh need to have some sort of
memory by default U so just standard
artificial uh recurrent networks and
leaky integrated fire neurons neither of
them have this long shortterm memory so
in the uh deep Learning Community people
tended to use this model called
LSM uh which was really really powerful
because it had this specific mechanism
for storing and updating a me something
called a memory cell so it could U gate
the input coming into the memory cell it

could uh decide to forget things it
could also gate the things going out of
the memory cell and this really helped
uh this model to work really work
have have long memory essentially long
short-term
memory and so we wanted to do something
very similar for spiking neural networks
because leaky integrated fire neurons by
default do not have uh this memory and
so one of the interesting observations
uh that we had was that in
Neuroscience the activity of the neurons
don't really correspond to like a very
typical leaky integrated fire Neuron
model pi because there seems to be some
sort of adaptation happening so if
the neuron has constant input so I mean
this is this is kind of an example where
if if you give the neuron constant input
for leaky integrated fire neuron you
mostly expect the spikes to be evenly
spaced U because every time it hits the
threshold uh the neuron spikes and then
it's reset to a lower value and then the
input kind of drives the mem brain
potential up again it spikes again and
so on but what you actually see in
biology is some sort of adaptation so if
a neuron spikes then after every Spike
the next Spike uh is pushed back it it
spikes less for a while and then after
that uh it spikes again and then again
it spikes less for a while and SW on
there's kind of like adaptation of the
Dynamics uh and you see this everywhere
so in the pyramidal cells in the mouse
and in the human neocortex so this is

data from the Allen Institute where they kind of characterize cells from various different animals and um so this kind of measure of this adaptation is called an adaptation index and essentially what you see is there's a big chunk of neurons in uh humans and mice which are which have an adaptation index that is large which means that there is this adaptation happening and so what we did uh was to try to model this uh and include this F and leaky integrated neurons and we came up with a model called uh an Adaptive leaky integrated INF fire neuron where the idea was quite simple it was uh to adapt the threshold every time a spike happened so every time a spike happened the threshold would go up a little bit and then DEC back to the base value and so as it turned out this simple mechanism so this this kind of threshold adaptation was enough to uh put the performance of spiking neurons on like very very close to lstms when trained with back propagation through time so this is a plot where you see this uh so this was uh tested on mnist a sequential Mist where you give the uh input one pixel at a time to the recurrent Network and then it actually to classify what the digit was at the end and uh so just standard leaky integrated fire neurons does really really poorly so it has like 50% test accuracy whereas just adding this adaptive threshold which is the model that we called LSN gets the performance

up to 93.7 right and lstms are up here

at

98% and um so with with some other

additions uh and using parameter

sparsity uh it goes up to 96.4 right and

this is with spiking neurons so it turns

out that uh so this this was like one

one example of like using like

biological data and basing the model on

that and seeing a huge Improvement in

what the models could

do okay uh so how am I doing on time uh

uh so we're about halfway through the

session now so how how long do you much

more do you think you'd

like um probably about 15 more minutes

is that okay we're halfway through the

hour H you can take an extra 15 more

minutes it means we'll have sort of 15

minutes for chat at the end so it's up

to you kind of how you want to uh divide

it I would like to remind everybody if

questions they can kind of jump in as we

go as well there's it's it's really

dense this one I'm loving it but like I

I guess yeah the timing we might have a

little bit less time for chat at the end

but if people want to stay on um the

call like I can stay on as well for a

bit longer if you can as well and and

yeah absolutely sure I mean yeah so I

will I will try to like uh do speed it

up a little bit at least like skip some

parts and then you can go and look up

the paper right okay don't go all the

details okay but uh so uh so we kind of

also had um um models that were able to

take advantage of this temporal coding

that I mentioned actually using time to do computation and uh so to kind of quickly uh summarized this the idea was that so in the brain uh you kind of have a specific mechanism that controls the speed of transmission of signals uh so this is called um um so there's this thing called a myelin sheath and there's there's been like a lot of recent evidence that uh so the myelin sheath is has been known to control the speed of the signal transmission there's been a lot of recent evidence that the myelin sheath actually adapts in adults which means that that seems to be like an additional dimension of learning that happens in the biological brain in adult brains and um so we wanted to kind of see what can be done with just controlling the speed of transmission so which in a computational model translates to a delay so a delay in communication between one neuron and another neuron and so what we tried to do was to test out uh if we could train a model with only synaptic delays uh without training the weights at all and see how far we could go on deep learning benchmarks right and so we did this and uh so we had like a temporal encoding and we constrained the weights to only three values so some so it was something like -10 , 0 and $+10$ uh and the weights were randomly initialized and then fixed and then we trained only the delays of the uh Network synaptic delays and it was quite interesting to see that

just by training delays you can get performance that's comparable to like your standard uh both both uh standard artificial neural networks uh without convolutions and uh standard spiking neural networks uh where the weights are trint can I can I just ask is that with the Adaptive neurons as well that you were just talking about or is this just with the delay uh so this does have some form of adaptation in the neuron uh so it's actually a slightly different form but in this particular case uh we were working with feed forward networks and not tricker networks okay so uh the memory didn't matter so much okay thank MH so this is an example of like using uh time uh for computation and we there's ongoing work to kind of extend this even further and to see really how far we can uh push using time as to compute because so what is interesting is uh when when you use delays for example uh in a computational model if it's implemented properly in appropriate Hardware a delay is just uh a neuron not doing anything until a certain point right which means that when you're when it's not really doing anything when it's not sending um message to the following neurons you're not consuming any energy and so this is a very uh nice way to make the models more efficient by tra by having delays and uh by using by kind of like reducing the activity of the overall

Network

okay

so uh so so far we've been like looking at at spiking neural network so which were these models that were based kind of like very very closely based on biological Dynamics and uh still using all of these principles of sparsity and event based communication and temporal coding but spiking neural networks of this particular form still don't work um as well as the best deep uh Network models so essentially in 2018 lstms were when when we were doing this adap neurons lms were the dominant model but then as we as I mentioned in the beginning of the talk uh there have been like newer models that have been performing even even better and so now we really have these Transformers and related models that do really really well and scale really really well and spiking neural networks pretty soon kind of hit a uh ceiling in terms of how well they could do and we had to think about uh ways to push the effectiveness of these models beyond that while keeping all the advantages in terms of sparsity and efficiency so we started thinking about uh really using these principles biological principles and designing models uh from first principles uh to take advantage of sparsity and event driven Computing and so on but then without really modeling the neuron directly based on biology uh and the I mean in for the sparsity in time the basic idea really

was that you had like this threshold gating function uh that would decide when communication went uh to other neurons and when it did not and we realized that we could just put this use this for other uh architectures that tended to work really really well for things like language modeling so uh and and with both uh these models and with spiking Dr networks what happens is that uh because you have this uh activity sparsity which is essentially that not all the units are active all the time when you train these models the backward pass is also sparse because you only really need to propagate the gradients through the uh units that were active and a few other units because of the particular shape of the pseudo derivative that we chose but you do not have to propagate the gradients through all the units and if you can actually take have some Hardware that can take advantage of this you end up getting like way more efficient training as well so not just the forward bus so this is kind of uh uh an illustration of this so the forward pass over here with these blue arrows is uh only goes through units that are active and the backward pass uh when you're calculating the gradients again only goes to the units that are active and a couple of other units uh that were close to being active but most of the other units are skipped entirely in this computation and we decided to to take this model called a gated recurrent unit

which is another recurrent Network model that works uh quite well I mean it's still lot on par with Transformers but uh it works uh so mean it was it was one of these models that had like a really quantitative advantage over leaky integrated Fire based spiking networks and we wanted to make it activity sparse as well and uh so we took the Gru and added a even generating mechanism to it using a gate and uh few a few other features like having a reset and having um refractory period and so on and with this model we were actually quite easily able to match the performance of lstms or or like the best language models recurrent language models u in terms of perplexity so that's that's kind of like what you see over here so there are like two different data sets one is the penry penry bank and one is the wiktex 2 and the uh bold values are like the best recurrent Network model performance on these two data sets and with our uh event based Gru model as we called it we were able to get um perplexities that were very very close to these um recurrent models but we had extremely high activity sparcity right so if you look at just maybe this one example uh with the egru on the penry bank we had about 80 85% activity sparcity and so if you had had Hardware that could actually take advantage of it it would uh it would use like 85% less computation if the hardware could actually take full advantage of it and the backwards parity

was close to 50% so you really end up doing like 50% less computation for training the same model right and we were also able to use like way fewer parameters uh to do this so overall uh these models I mean so the thing is this these research that we showed uh all the training was done on gpus because we actually don't have the hardware that can take advantage of all of this sparsity uh but this is uh in some sense like calculation of exactly how much energy we would be able to save if we had this

hardware and U so we we were we were also kind of able to tune uh kind of tradeoff uh sparsity with performance within certain limit so you see here that we don't even though we are very close to the recurrent model uh performance we don't actually reach it and then there is a way to kind of make it less sparse and reach this uh close and you still kind of end up with some Energy

savings uh so for example uh so we actually implemented this on a neuromorphic chip uh recently uh is a neuromorphic chip called Spiner 2 which is developed by uh some folks at T Reston in Germany and what was kind of interesting is uh in if you if the energy consumption for single batch uh for Spiner 2 which is this green bar over here as you can see is significantly lower than running it on a GPU so uh this blue bar is the a100 GPU which is like the one of the more recent

ones and the orange one is like a slightly older GPU so it's like a little bit slower and uh Spiner 2 and this was like actually a very uh Proto uh the implementation was essentially a prototype so we were not actually making full use of the chip but uh it was able to make use of some of the sparsity um on the device and you kind of get like a huge energy Advantage even with that although for larger batches you don't right so this is kind of still one of the uh difficult areas of neuromorphic computing where it's not able to take advantage of lar larger batches so um yeah so this is kind of uh on archive you can go look look it up uh but uh so we kind of like getting there slowly um getting getting all of the sparsities on Hardware okay so we can also use sparsity to make training very efficient I won't go into this in detail but essentially you can use algorithms that so far were considered completely uh computationally prohibitive uh in terms of the computational expense you can actually that so these algorithms uh become more practical if there is activity sparsity and parameter sparsity okay so so and we we have explored like a really really really tiny part of the properties that can make model scalable which is just activity sparsity and parameter sparsity and we have mostly explored it only in terms of algorithms and so there's obviously like a long way to go before

we start reaching this scaling uh of biological uh substrates and biological algorithms but we're kind of like trying to move towards this direction so that's kind of like the conclusion ution which is that um sparsity and locality are essential for scaling up models due to energy constraints and there might there are also other clever ways to use physical substrates uh to use to improve Energy Efficiency so using time to compute communication and others and the next step is to really like try to scale it up and start to catch up with uh Transformer based models on tasks that are a lot more closer to what people use in the real world so a lot of this work was done with lots of other people uh so this is the group that uh group and collaborators that we have we have like a large group in ruer University bohomo so you might you some of you might have actually seen a talk by David Capel who gave a talk at mream a while ago so uh and we have a few people from T Tron and uh there's is the group that actually develops the Spinner to hardware and makes all of this possible so thank you very much um yeah hopefully we have some time for discussions thank you so much that was absolutely fascinating there's a lot to unpack there so Harris you've got your hand up if you want to ask a question yeah ER I just wanted to uh say Kudos it was very interesting and I have a couple of uh questions related to this presentation uh first of all H you you poed one um very interesting question

that has to do uh with the following um
can we scale even better let's say
moving effectively moving beyond
Transformers and as far as I understand
it uh because uh these large language
models the llms are very are all the
hype these days so as far as I
understand it not all llms are are
Transformers
uh but the other way around is true and
the reason for for that H is because uh
there are some um llms that are not
based in in
Transformers as you mentioned uh you
mention the recurrent neural networks of
course and that's one one type let's say
and uh another uh uh subtype that is is
based on rnet is
the long shortterm memory networks that
you are group is is actually
investigating and uh what I wanted to
say is
that trying to uh put everything
together is there a
possibility for moving beyond uh the
Transformers and scaling even better by
opting for what is called poly
Computing and and I mean poly Computing
that is capable to uh to combine uh
different types of computing in a multi
Computing Paradigm like classical
Computing Quantum comp Computing
neuromorphic and even Optical so what do
you think about
that yeah that's a very interesting
question I mean so there's definitely
like a lot of potential in these
unconventional Computing devices that

various people are developing right and each of them have different strengths and different weaknesses so it would make sense to combine them together in a way that uh we use the strengths of like some of them without uh being affected by their weakness so much but uh well I mean at least in terms of efficiency itself if using like multiple devices or multiple different uh Computing nodes always kind of leads to this problem where you do a lot of communication and communication is expensive right so even on a CPU or a GPU or like most Computing devices that we have today uh the communication is really the bottleneck so that's probably uh I mean that's actually like one of the main reasons why we are looking at activity sparsity because that's that has to do with communication so communication tends to take a lot of energy it consumes a lot of time and so uh when with like multiple different types of commun computing the main problem that would have that would need to be solved is how do you make communication efficient both like on a physical substrate you know like both both in terms of like the device itself and also in terms of algorithms okay thank you thank you very much um yeah I had a couple of questions as well I think it's it's fascinating um that you're saying essentially that you can get get the same performance of the

Istms but like with about 80% less power like I think that's really amazing do you think that this these types of architectures will be used sort of in the near future to make say gpus more efficient or or implemented in some specific use cases maybe where sort of the sort of temporal sparsity is particularly important the temporal coding is particularly important so I'm quite interested to hear your take console what the applications where you see this being used in in the near future MH so the temporal coding I mean I've not really seen uh much attempts to make use of it I think it's it's probably like one of those things which is like the hardest to make use of both both from algorithms and Hardware point of view but people are trying to make use of uh this activity sparsity so Nvidia for example recently uh added support for a certain type of sparsity like it's still like very very limited but uh they started adding support for a certain type of sparsity which in principle could be used for activities parity quite well and there have also been like a couple of deep learning accelerator companies that uh do activities or I mean so okay I mean so actually like the I mean there's probably like a bit of context that's required so the reason that activity sparsity is really hard to make use of in Hardware is because it's Dynamic right so if you have parameter sparsity and if the parameter sparsity is

fixed you kind of know which uh particular weights you need to load at what times and which ones you don't but if this is changing all the time then it's really really hard to optimize uh its use and so this Dynamic sparcity is what is like extremely challenging for gpus and like most conventional devices so uh nvidia's attempt at adding this support for sparcity is to to some extent address uh being able to use Dynamic sparcity although in's case you cannot use uh I mean so basically using even static spity is very very hard unless it's designed in like a very specific way uh

so and and so this deep learning accelerator company that I was mentioning so they really have done they've designed a chip completely from scratch so it's a company called cerebras which can make use of Dynamics parcity and uh and it's also like super scalable because essentially what they've done is created like this huge chip on a single wafer which means that the communication overhead is much much lower and uh it works like really really great for training like extremely large models if if you have the right kind of sparcity on them right so it's still not like completely general purpose it's still like very closely tied to Transformer architectures but it can use activity sparcity really well uh if implemented correctly but uh they don't really go beyond that so they don't do

like time based Computing or make use of
like any of the other aspects especially
asynchronous Compu Computing for example
is as far as I know only
neuromorphic people working on
neuromorphic devices seem to even use it
although well I mean I guess like
somewhat ironically we haven't fully
figured out how to use it
effectively interest I was actually just
going to ask you a bit about Dynamics
vast because I'm in interested in um
sort of neuro neur taking inspiration
from neurod development and I I think
it's really interesting that in the
developing brain like say from about Age
2 to 10 you have huge amounts of
synaptic pruning like you actually lose
about 50% of your synapses and that's
essentially a a key component of how we
learn um so before I saw this talk I was
thinking about how if anyone's actually
taking inspiration from neurod
development for example thinking about
sparsity and how the brain sort of
develops by pruning away all the sort of
unwanted connections is is that which is
people are thinking about does that fit
into what you were just saying a little
bit about Dynamics B here is that
slightly
different uh no I mean it's actually
very very closely related and uh I mean
we've been like thinking about it a
little bit also but so essentially uh
what we have come to realize is that so
there are two
different aspects to having a model that

performs well so one is just the inference so if you have sparsity uh you can make predictions with this model or like kind of run the model forward uh and uh any sort of sparsity helps it makes it very very efficient but this does not necessarily make the model easy to train right so you need to have there is this like really orthogonal uh concern where you need to be able to train the model well and it's little bit in conflict with having sparsity so like one of the well at least like from a very very non-expert perspective what it seems like is this neuro development uh this kind of like pruning that happens it seems like there might be like a learning phase where a certain type of learning happens that is a lot harder to do if the overall network was really sparse and then you have this like huge pruning space face uh and then you have like a different network that learns that can still learn but like a lot more slowly perhaps or but but it's like a lot more energy efficient it interesting because yeah like the brain although it's pruning it's also growing and and the part of the reason why it's growing so much is because the myelination which is interesting because you talk about that in terms of the synaptic delays so maybe sort of there's some balance between the sparsity pruning and then sort of the delay maturation that's introduced somewhere in a dynamic way um that's

important to do I don't know yeah I mean
I think uh I mean yeah so I mean
obviously like in biology uh you know
like specific things don't just do one
thing right they do like a number of
different things so that's that's the
case with Malin as well for sure uh but
uh so kind of what what is interesting
is that with delays I mean if you can
control the delays and if you can work
with longer delays then you actually
need less activity in the network
because uh I mean so if you take like a
network with like absolutely no
long-term memory so like Ley integrated
F neuron or a v Network and you kind of
try to train this network to store
something for a longer time one of the
ways that it can discover to store
things is literally uh persistent
activity so you have like one neuron
that keeps firing all the time because
it's storing one particular bit of
information and then at the end if this
neuron is firing you know that bit is
stored and if not it's not right so it's
kind of I mean it doesn't always emerge
and it's pretty hard there's a reason
why these R networks don't work well for
like long-term memory but uh if you if
you make the problem simple enough you
do see this right and in when you train
like a spiking Network or an lstm you do
see like persistent activity on and off
for various things and if you had delays
you could just completely avoid this
right you could just like uh say this
neuron will will send the message when

it's ready after like a few time steps
or so and only then the other neuron
needs to process this
information okay interesting that might
touch on my which I'll make my final
question because we have to drop and
everyone's dropping off absolutely
unshare your screen sorry I should have
probably asked to do that earlier um I'm
I'm quite interested in this idea of so
artificial neural networks they have the
sort of this like always on an
activation function whereas people
contrast that with spiking neuron
networks which are like neurons and like
they're on or they're off and what's
interesting I guess from a neuroscience
perspective is because a lot of the
brain is actually gl cells which
actually do have more this continuous
Activation so the brain is actually
quite a hybrid between more like Anns
and snns essentially so I'm super
interested in sort of if if you think um
like maybe the future will be um hybrid
in some way um oh Carl Frist has entered
the way waiting room I think I'll ad
admit him I don't know if he's in the
right
meeting we're meant to have an sa
meeting
okay maybe yeah so that was my last and
final question sort of do you think the
hybrid system um would uh be something
which people are interested in or
looking at in the future or is that not
something that I see uh that's an
interesting question

so well I mean the question I mean so I think the related question that needs to be answered is do hybrid systems really provide any sort of advantage in terms of uh computational Power or any any other thing and at least like as far as I know there is no reason I mean so if you can get like this uh event based system to work and at least like theoretically it's they're like equivalent they're supposed to be equivalent to artificial like you know non-event based systems then there wouldn't be any reason to use a hybrid system but it could very well be the case that there are certain constraints where just an event based system would be a way to slow or you know so that that might be like a concern so it just like takes too long to reach decisions or something like that and then and then you might need like hybrid systems as well yeah I mean so I'm happy to discuss I mean if you're interested in this area I'm happy to discuss with you offline as well I'm super interested because as well I like the sort of you get with the gal cells there actually some electrical communication they have Gap Junctions and they have these sort of slow oscillations which almost like event based oscillations almost take advantage of or ride so there's sort of an embedding nesting there of the the uh spiking and non-spiking so um yeah it's something I'm super interested I don't I just yeah I don't know if the Silicon world is as interested in it as me

though as a
neuroscientist yeah but I'm happy to ch
to it
offl okay well thank you so much that
was absolutely fascinating um I'm sure
people might want to reach out to you
with some questions s of offline because
this will be put on YouTube so if they
do what would be the best way for them
to contact you if that's okay yeah
absolutely it's okay I mean I have my
contact information on my website okay
great so email is probably like the best
way to contact me cool okay right thank
you so much and that's everything that I
have to cover um yeah thank you thank
you very much for inviting me to the
talk it was it was very interesting yeah
amazing yeah I'd love to stay in touch
and chat some more about this as well so
thank you so much
absolutely all right see you
byebye